

A three layer stacked multimodal transfer learning approach for deep feature extraction from Chest Radiographic images for the classification of COVID-19

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ABSTRACT

COVID-19 has had a profound impact on global health, targeting the human respiratory system and causing significant disruptions to human life. To aid in effective diagnosis and classification, numerous machine learning and deep learning models have been developed to analyse limited Chest Radiographic and Computed Tomography images. In this study, we propose a three-layer stacked multimodal approach for deep feature extraction from a high volume of COVID-19 Chest Radiographic images. The proposed model utilizes eight transfer learning models pre-trained on the ImageNet dataset, evaluated based on key performance metrics such as accuracy, precision, and recall. The unique stacking model integrates the outputs of these transfer learning models, extracting deep features through a three-layer architecture. These features are flattened, concatenated, and passed through seven deep dense layers with varying kernel and bias dimensions to achieve optimal classification performance. The approach was applied to a large COVID-19 Chest Radiographic images dataset, achieving the highest accuracy (95.79%), precision (95.44%), and recall (96.65%) when compared to state-of-the-art models. This study demonstrates the effectiveness of leveraging a stacked transfer learning multimodal framework for COVID-19 diagnosis. The proposed method not only ensures high accuracy but also provides a computationally efficient solution for analysing large-scale radiographic datasets, positioning it as a robust tool for aiding in the early detection and classification of COVID-19.

1. Introduction

Some people contracted the illness in a nearby market due to the outbreak of an unidentified sickness in China towards the end of 2019. When the illness first emerged, no one knew what it was, but experts soon discovered that its symptoms resembled those of flu and coronavirus infection (Chen et al., 2020; Huang et al., 2020; Lima, 2020; Struyf et al., 2022). This widespread sickness precise cause was previously unknown. Still, after laboratory testing and real-time Polymerase chain reaction (PCR) investigation of positive sputum, the viral infection was verified and given the name "COVID19" on the advice of the World Health Organization (WHO). The COVID19 pandemic quickly expanded across international borders, having a disastrous impact on the population's health, welfare, and economic situation (Liao et al., 2020; Ghaderzadeh and Asadi, 2021). Millions of people were affected by the global health catastrophe brought on by the coronavirus outbreak in December 2019, which also had a severe impact on the international

economy (Khan et al., 2023). As of April 2022, coronavirus had affected more than 507 million individuals worldwide and caused more than 6 million fatalities (Aslani and Jacob, 2023). For the majority of persons infected, the COVID19 symptoms brought on by the novel disease Severe Acute Respiratory Syndrome Coronavirus-2 (SARS-CoV-2) are difficult to distinguish from those of other typical infections (Banerjee et al., 2020). The overall white blood cell count of COVID-19 patients was reported to have declined marginally, as with many other disorders, but the lymphocyte count of these individuals was reported to have decreased dramatically (Wang et al., 2020; Zheng et al., 2020a). Further research is required because there is a dearth of data on early risk factors for severe cases (Zheng et al., 2020b; Huyut, 2023; Qiao et al., 2021; Zhang et al., 2022). Globally, COVID19 has had a terrible influence on people's life. The human respiratory system is attacked by COVID19, disrupting human health. COVID19 may occasionally result in death (Najman et al., 2020). There is a slowdown in the economies of various nations. In order to cut costs, many businesses must fire employees

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(Toharudin et al., 2020). As they must study at home, students’ social activities are disturbed, and their academic performance is subpar (Cenggoro and Pardamean, 2023). In addition to helping doctors identify and cure disorders, medical imaging objective is to reveal the human anatomy. Disease detection like Lung cancer and COVID19, radiological quantitative image analysis is an important prognostic component. In order to diagnose, stage, and monitor patients, for instance, CT scan is a crucial test (Kollias et al., 2023). It is advised that radiography be used instead of a CT scan whenever medical imaging is required because of issues including the possibility of disease transmission by means of a CT modality and its high cost, which can result in major consequences for the patient and healthcare systems (Amis et al., 2007). X-ray imaging is far more thorough and affordable than traditional investigative procedures. The diagnosis process is completed extremely rapidly since transmitting an X-ray digital image does not involve moving from the access point to the inquiry site. The medical triage of patients using chest radiography is quick and convenient (Ghaderzadeh and Asadi, 2021).

The past few years have seen tremendous advancement in machine learning. For example, it is used in medicine to classify cardiovascular illnesses, treat diabetic retinopathy, and other things (Alghamdi et al., 2020; Hammad et al., 2018; Khalil et al., 2019). Medical professionals can now employ the Convolutional Neural Network (CNN) for a variety of tasks, including the diagnosis of skin lesions, brain tumours, and breast cancer (Cenggoro and Pardamean, 2023; Esteva et al., 2017). Deep learning (DL) methods have been successful in quickly analyzing and delineating infectious areas in Radiological Images (RI) (Khan et al., 2023). Deep Neural Networks are being used in new methods to solve medical imaging issues, such as lung pneumonia region segmentation and automatic identification (Kollias et al., 2023).

1.1. Contribution and highlights of the research work

- The aim of the proposed work is to deal with the limitations of the existing work using a three-layer stacked transfer learning multi-modal to perform critical analysis of high-volume Chest CT images for effective diagnosis and classification into Covid-positive and Covid-negative class.
- In first layer of the stacking model, a variety of transfer deep neural network (VGG-16, VGG-19, ResNet50, ResNet101, ResNet152, InceptionV3, InceptionResNet152, Xception, DenseNet121, DenseNet169, DenseNet201, and MobileNetV2) used for deep feature extractions from CT X-Ray Covid-19 images. These deep features were flattened and concatenated by applying variety of hyper parameters such as adaptive learning rate, binary cross entropy, log loss and Adam optimizer.
- The extracted deep features from first layer were then fed to a fine grained Deep Neural Network (DNN) models (VGG-16, VGG-19, ResNet50, InceptionV3, InceptionResNet152, DenseNet121 and MobileNetV2). The neural network models proposed in the second layer used for deep feature extractions by applying more fine-tuned and grained hyper parameters.
- The deep features extracted by the deep neural models in the second layer were then integrated and passed to deep dense neural network model of seven layers to obtain accurate and effective classification of Covid-19 class. Each layer of seven layers deep dense model has used kernel and bias of varying dimensions.

Abbreviations

CT	Computed Tomography
CXR	Chest Radiographic
PCR	Polymerase chain reaction
WHO	World Health Organization
SARS-CoV-2	Severe Acute Respiratory Syndrome Coronavirus-2
CNN	Convolutional Neural Network
DL	Deep learning

(continued on next column)

(continued)

RI	Radiological Images
DNN	Deep Neural Network
AI	Artificial Intelligence
TCNN	twin Convolutional Neural Network
ILSVRC	ImageNet Large Scale Visual Recognition Challenge
FLC	fully connected layer

The organization of the study is as follows: Literature survey discussed in Section 2; Materials and Methods of the proposed work is shown in Section 3; Results are discussed in Section 4; and finally, the conclusion is presented in Section 5.

2. Literature review

Apostolopoulos et al. technique’s for COVID-19 identification was one of the first Artificial Intelligence (AI) based methods (Apostolopoulos and Mpesiana, 2020). The authors employed VGG, MobileNetV2, Inception, Xception, and Inception ResNetV2 (Howard et al., 2017; Xia et al., 2017; Chollet, 2017; Szegedy et al., 2017) as modern CNNs in this investigation. Using transfer learning, the COVID-19 was diagnosed. DL analysis frequently employs the transfer learning technique, which allows computer models that have previously been trained for a particular task (such as the categorization of two picture classes, such as dogs and cats), to reuse the information they have learned and applied it to a different, interrelated task (e.g., classification of COVID19 and non-COVID19 CXRs). Apostolopoulos et al. (Apostolopoulos and Mpesiana, 2020) found that the three-group (normal, bacterial pneumonia, COVID19) and two-group (normal, COVID19) classifications had the highest accuracy rates at 93.48% and 98.75%, respectively. The suggested model was further evaluated on a different dataset that had additional instances of viral pneumonia; the findings for three-group and two-group classification were stated to be at 96.78% and 94.72%, respectively.

A twin Convolutional Neural Network (TCNN) architecture was suggested by Rahimzadeh et al. (Rahimzadeh and Attar, 2020) to extract parallel deep features from an image with two popular CNNs, Xception and ResNet (Chollet, 2017; Lin and Jegelka, 2018). The constant model output or response at the final node or layer is referred to as a dense or latent feature. The final forecast was created by combining the extracted latent features. 99.05% and 94.40%, respectively, of the classification tasks using two and three groups were accurate. According to the findings, a TCNN strategy might improve algorithmic COVID-19 prediction ability. A TCNN with shared layers called CVDNet was proposed by Ouchicha et al. (2020) and Rahimzadeh et al. (Rahimzadeh and Attar, 2020) in a manner similar to theirs. The model was evaluated on three classes classification and achieved 96.69% accuracy. For natural image categorization, SqueezeNet is a skill full CNN that utilize less parameters than the other models (Wu et al., 2019). Ucar et al., (Ucar and Korkmaz, 2020), created a new network named COVIDiagnosis-Net using the SqueezeNet building blocks.

(Tang et al., 2020) states that after normalizing the images, enhanced attributes were extracted from them and fed into deep learning-based image categorization systems. Five cutting-edge CNN systems, including VGG19, MobileNetV2, Inception, Xception, and Inception-ResNetV2, were examined for COVID19 diagnosis in control and pneumonia pictures under a transfer-learning scenario. The first trial used the COVID19 data and previously collected normal data, but it also contains 714 cases of bacterial and viral pneumonia. The second experiment made use of 504 images of controls, 700 images of bacterial pneumonia, and 224 COVID19. With accuracy value of 0.9678 and 0.9472 for the 2- and 3-class classifications, respectively, the MobileNetV2 performed the best. Anum Shafiq et al. (2022), estimated COVID-19 mortality rates in Italy through the application of artificial neural network modeling and maximum likelihood estimation. The findings revealed that the neural network model, employing nine neurons in its hidden layer, exhibited

exceptional predictive accuracy, with a maximum deviation of -0.14% and an R value of 0.99836. T. N. Sindhu et al. (2024), employed the Burr-Hatke exponential model to evaluate the reliability and survival characteristics using maximum likelihood estimation and artificial neural networks. Results demonstrated that ANN models achieved high predictive accuracy and provided deeper insights into reliability metrics and survival properties.

Automated Machine Learning (AutoML) has emerged as a promising tool in diverse fields for its ability to streamline complex machine-learning processes and reduce the need for manual intervention. Recent studies have highlighted AutoML's potential in medical imaging for hyper-parameter optimization, neural architecture search, and data augmentation, addressing challenges like modality-specific characteristics and parameter tuning. AutoML has also been applied to diagnostic systems for adsorption cooling and desalination facilities, demonstrating efficiency in failure prediction and operational optimization. In the energy sector, AutoML techniques have proven effective in optimizing adsorption cooling systems and improving sustainable design. Furthermore, in the Industrial Internet of Things (IIoT), AutoML models have significantly enhanced predictive maintenance by reducing computational overhead while achieving high accuracy. These advancements underline AutoML's versatility and its growing impact across multiple domains (Ali et al., 2024; Krzywanski et al., 2024a, 2024b, 2024c; Hadi et al., 2023).

AI-based algorithms, particularly the Data Ensemble Refinement Greedy Algorithm (DERGA), in predicting critical health outcomes with high accuracy. In cardiovascular disease, DERGA identified six key predictors, including Platelet-derived Microvesicles, achieving a prediction accuracy of 98.64%. Similarly, in COVID-19 patients, DERGA and artificial neural networks (ANNs) effectively utilized hematological indicators and complement-related genetic variants to predict ICU admission, morbidity, and mortality with accuracies exceeding 97%. These studies underscore the utility of AI-based models in optimizing clinical decision-making, reducing resource use, and providing early risk assessment by identifying optimal predictive patterns of clinical, hematological, and genetic parameters (Asteris et al., 2022, 2024a, 2024b, 2024c). Safe ML which is another major aspect focuses on developing statistical methods and metrics to assess and mitigate risks, ensuring reliability, fairness, and transparency in AI applications discussed to ensure for better performance of the models (Babaei et al., 2025; Giudici, 2024).

2.1. Research gaps

The diagnosis of COVID19 and related lung diseases by the state-of-the-art existing models have limitations in many ways.

- (i) The datasets used by the earlier models were limited in size; only very few COVID19 images were taken into consideration.
- (ii) Most of the datasets used by the earlier models considering three classes of disease namely: normal, bacterial-pneumonia and few covid-19 images.
- (iii) Earlier, State-of-the-art existing models focussed either binary classes for Covid or non-Covid classes or into three different classes i.e. normal, bacterial-pneumonia and Covid-19 classes using a single deep neural network models.
- (iv) Some authors have used deep feature extractions from CT X-Ray images by using limited capable models based on Twin CNN, ResNet and Xception and combined the latent features for final prediction.
- (v) some author has used Twin CNN with shared layers CVDNet model for three class classification.

Table 1

Dataset Distribution in COVID19 CXR.

Type	COVID19 Negative	COVID19 Positive	Total
Train	12576	14411	26987
Test	1416	1583	2999

Images were resized into 224x224 for training the models.

3. Materials and Methods

3.1. Dataset description

The COVID19 CXR dataset constructed from multiple different publicly available data sources was used for the experiment. This dataset contains two classes of COVID19 positive and negative chest radiography images as shown in Table 1.

3.2. System requirements

Google Colab Framework was used to conduct the experiment having 22 GB RAM and Storage space of 166 GB. The dataset was divided into 9:1 ratio for training and testing. Different hyper parameters were used like loss function used was the binarycrossentropy, Adam optimizer in all the implemented models with 32 batch size. Each model was trained for ten epochs, each with 1350 iterations.

3.3. Method

This methodology contains two parts as shown in Fig. 1 data pre-processing and implementation of the various deep learning models. Firstly, the dataset was pre-processed by resizing the image dimensions from 1024x1024 to 224x224. Then the dataset was divided into two sets, the train set (90%) and the test set (10%). Secondly, the trainset containing Chest CT images was given to different deep-learning models to detect COVID19. Also, a novel three-layer stacking transfer learning multimodel was designed for COVID19 diagnosis utilizing CXR dataset.

After the training has been performed and the different models have been saved. Then the test set was given to these saved trained models to predict the COVID19 (positive and negative) output in the given CXR images.

3.3.1. Deep transfer learning models for the classification of COVID19

- I. **VGG16:** The ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2014 was won by the CNN model (Rezende et al., 2018) in two categories i.e., object detection and image classification on Image Net dataset with 94.2% test accuracy. It is considered the best model for image classification. It has 16 layers with the last two fully connected layers. The important part of the model is that instead of having large hyper parameters, it uses many convolution layers with 3x3 filters with stride 1. The pooling remains the same throughout the network with a max-pooling of 2x2 with stride 2. We have applied the VGG16 model to our dataset with additional layers of global average pooling of 2x2, dropout layer, batch normalization, and one dense layer for the final output. The total number of parameters was 14,847,297 out of which 132,097 were trainable. The Adam optimizer with a Learning rate of 0.001 and binary cross entropy as the loss function were utilized in the model (see Fig. 2).
- II. **VGG19:** It is a CNN proposed in the year 2014 by Simonyan and Zisserman (Sinha and El-Sharkawy, 2019; Karaci, 2022). It has total 19 layers with first 16 fully connected layers and last 3 dense layers. It was trained to categorize 1000 distinct items using the ImageNet dataset. Global max 2d pooling layer was added which gave the output of 512 feature maps. Other layers of dropout, dense layer, batch normalization, activation function and last 3

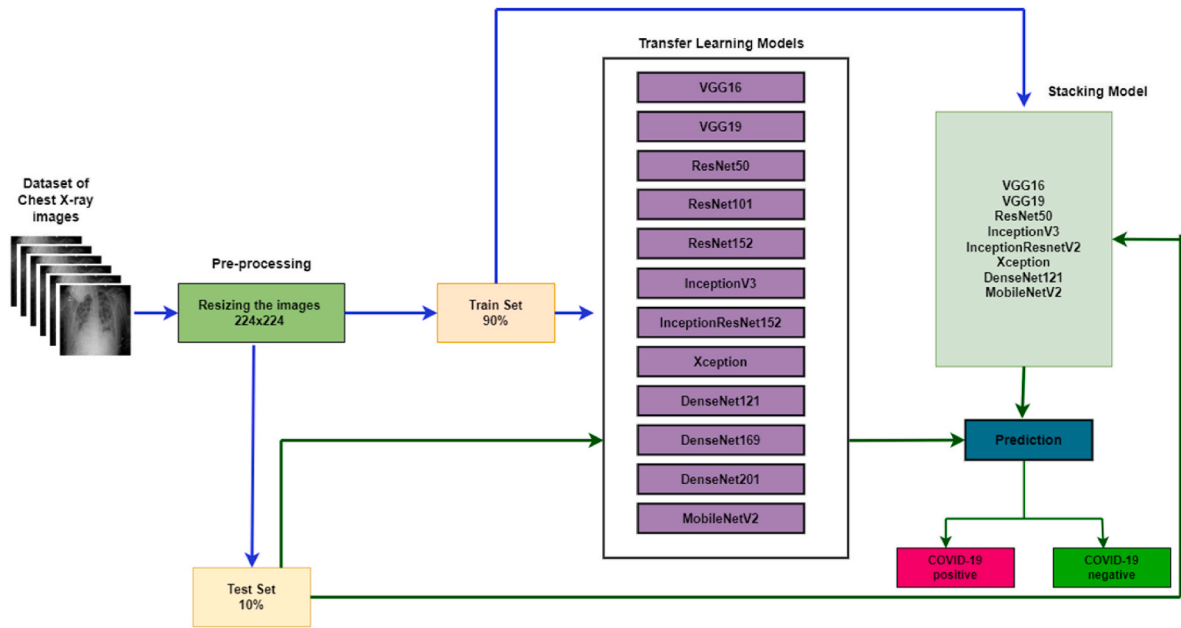


Fig. 1. Flow of the work for detecting COVID-19 diagnosis in CXR images.

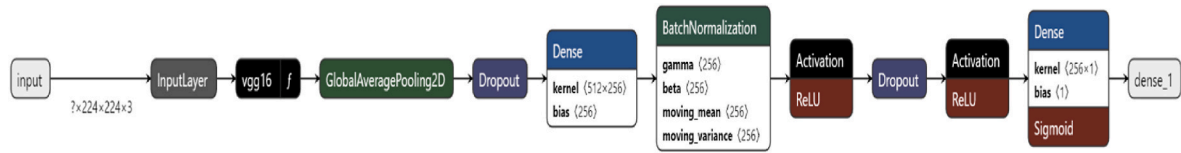


Fig. 2. Architecture of VGG16 model for COVID19 diagnosis.

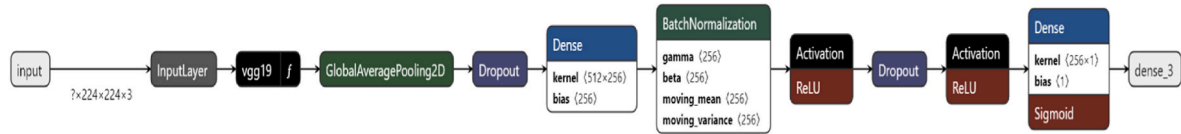


Fig. 3. Architecture of VGG19 model for COVID19 diagnosis in CXR images.

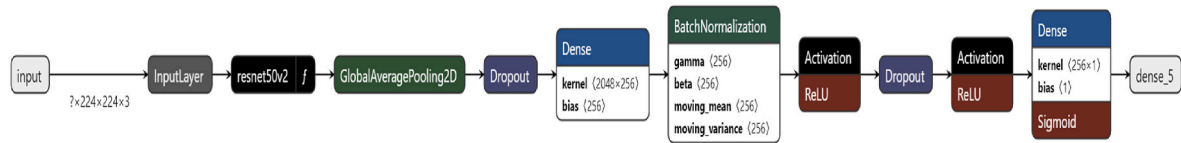


Fig. 4. Architecture of ResNet50V2 model for COVID19 diagnosis in CXR images.

dense layers. The total number of parameters was 20,156,993 out of which 132,097 were trainable (see Fig. 3).

III. **ResNet50V2:** It is another Deep Convolutional Neural Network that came into existence after the VGG16 and VGG19 (Wen et al., 2020). The ResNet50 has total 50 layers out of which 48 are convolutional layers, one is MaxPool layer, and one is average

pool layer. It contains stacking of residual blocks that has the feature of eliminating the problem of vanishing gradient that was not in the VGG16 and VGG19 models. The model was trained for ten epochs on the given dataset, which has a total of 24,090,625 trainable parameters. The hyper parameters (Adam optimizer

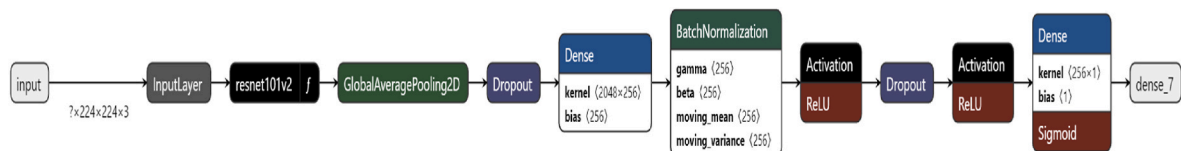


Fig. 5. Architecture of ResNet101V2 model for COVID19 diagnosis in CXR images.

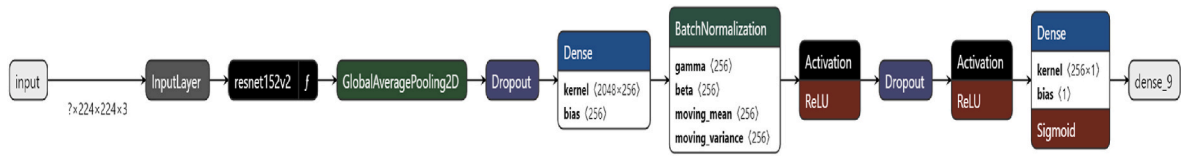


Fig. 6. Architecture of ResNet152V2 model for COVID-19 diagnosis in CXR images.

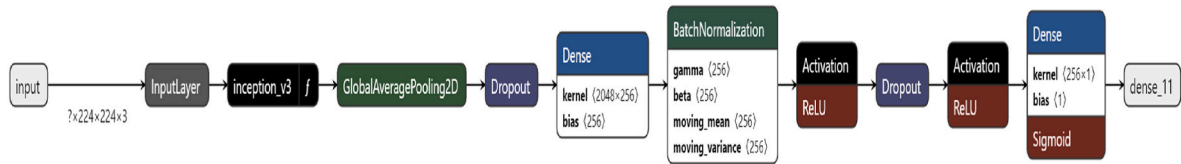


Fig. 7. Architecture of InceptionV3 model for COVID-19 diagnosis in CXR-2 images.

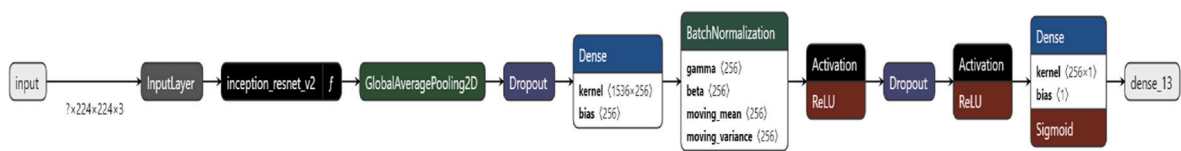


Fig. 8. Architecture of InceptionResNetV2 model for COVID-19 diagnosis in CXR images.

with a Learning rate of 4e-5 and binary cross entropy as the loss function) was used to train the model (see Fig. 4).

- IV. **ResNet101:** ResNet-101 has deep CNN (Hastomo, 2021) of 101-layer. The weights of the pre-trained version of the network on over a million of images from the ImageNet database can be loaded. The pre-trained network can categorize images into 1000 different objects, including keyboards, mouse, book, and other items. Adam optimizer with a Learning rate of 4e-5, and binary cross entropy as the loss function were utilized as the hyper parameter to train the model. The total number of trainable parameters was 525,313 during the model training (see Fig. 5).
- V. **ResNet152:** It is a deep residual network for image classification trained on ImageNet dataset (Hastomo, 2021). It has total 152 layers with 60 M parameters approximately. This model achieved above 95% accuracy in classifying the images. We applied this model on the given dataset with an image input of 224x224, estimates for a single complete model pass. The total number of parameters was 58,857,473 out of which 525,313 were trainable (see Fig. 6).
- VI. **InceptionV3:** It is an image recognition model that achieved 78.1% accuracy on the ImageNet dataset (Zhang et al., 2020). This model is the amalgamation of different ideas of different researchers. were employed Inception V3 model (Peng et al., 2022) utilized various strategies to optimize the network for enhanced model adaptability. Its network is big as compared to the Inception V1 and V2 models, yet its speed is unaffected. It is cost efficient. As regularizers, it employs secondary classifiers. InceptionV3 is better than the ResNet version because it eliminates the problem of exploding gradient in the ResNet model. The total no of parameters was 22,328,609 out of which 525,313

were trainable. ReLU activation function used in all the layers except the final layer. The final layer used sigmoid function for the classification of COVID19 (see Fig. 7).

- VII. **InceptionResNet:** A hybrid deep learning model (Endah and Shiddiq, 2020) has been applied to the COVID19 CXR dataset for the classification of Covid19 (positive and negative) classes. This model has a total of 164 layers and was pre-trained on the ImageNet dataset to classify more than 1000 objects. The weights of the pre-trained model was utilized to train the model on the given COVID19 CXR dataset. The total no of parameters was 54,731,489 out of which 394,241 were trainable (see Fig. 8).
- VIII. **Xception:** It is a deep CNN launched by Google and 1st runner-up in the ILSVRC, 2015 (Kumar et al., 2018). It is an extended version of the Inception module as it is better with enhanced deep separable convolution. The point-wise convolution and depth-wise convolution techniques used in this model. The point-wise convolution is 1x1 convolution (as in the inceptionv3 module) to change the dimension of the image. The $n \times n$ channel-wise spatial convolution is the depth-wise convolution. The Xception model is comparatively faster and gave higher accuracy on the ImageNet dataset than all other models with 525,313 trainable parameters on the given dataset (see Fig. 9).
- IX. **DenseNet121:** It is a dense network with 121 layers, of which 120 are dense layers and 4 average pool layers. All layers in the same deep dense block circulate their weights over the inputs which allow the deep layers to use the early extracted features. DenseNet is more efficient in using features and outperforms with fewer parameters. Using an Adam optimizer and a learning rate of 4e-5 hyperparameters, the network is trained across 10 epochs. The model's trainable parameters totalled 263,169 (see Fig. 10).

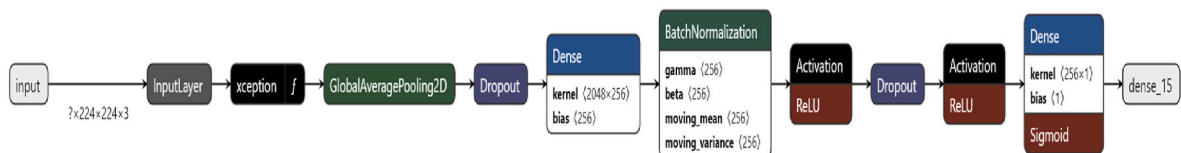


Fig. 9. Architecture of Xception model for COVID19 diagnosis in CXR images.

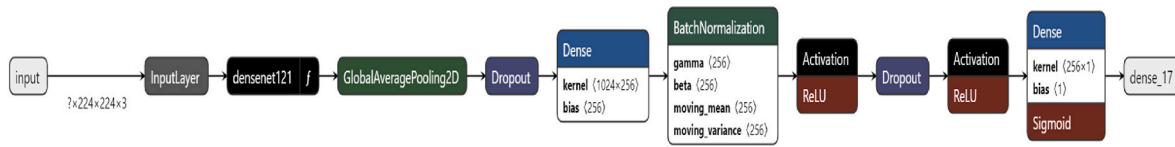


Fig. 10. Architecture of DenseNet121 model for COVID19 diagnosis in CXR images.

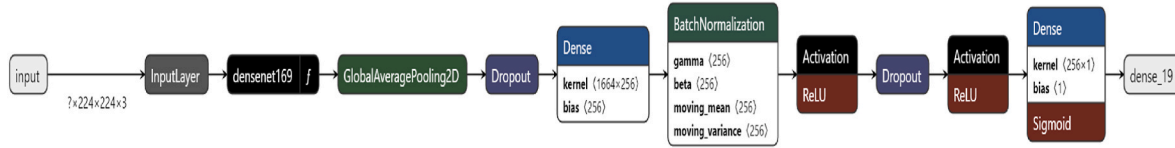


Fig. 11. Architecture of DenseNet169 model for COVID19 diagnosis in CXR images.

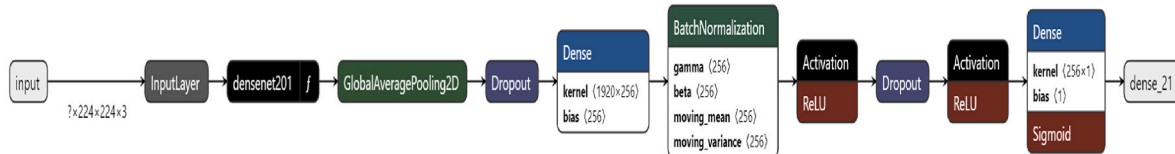


Fig. 12. Architecture of DenseNet201 model for COVID19 diagnosis in CXR images.

- X. **DenseNet169**: It consists of four dense blocks with 6, 12, 32, and 32 layers in each block. This network is denser and bigger than DenseNet121, with a 100 MB size. The last fully connected layer (FLC) was removed from the DenseNet-169 architecture, and as a result, a 256 node FLC, 128 node FLC, and finally, a 10 class FLC with Softmax activation for the output were constructed. ReLU activation function was used in all except the last layer. This model is trained for ten epochs on the given dataset with following hyperparameters: Adam optimizer with learning rate of $3e-5$, binary cross entropy for loss. The trainable parameters of the model were 427,009 (see Fig. 11).
- XI. **DenseNet201**: It is a deep CNN with 201 layers. It works on the idea that if there are short connections between the layers nearest to the input and output, convolutional networks can be trained to be remarkably deeper, more accurate, and much effective. It contains approx. of 20 M parameters and having a size of 77 MB. This model was trained on the given dataset for 10 epochs with Adam optimizer set to a learning rate of $3e-5$, binary cross entropy as the hyper parameters. The trainable parameters of this model on the given dataset were 492,545 (see Fig. 12).
- XII. **MobileNetV2**: It is a CNN with 53 layers (Howard et al., 2017). It is a more simplified, computationally less complex version 2 of the original MobileNet. Its inverted residual blocks set it apart from the MobileNet. It performs better on images larger than 32×32 and uses less parameters, making training faster. An excellent feature extractor for object recognition and segmentation is MobileNetV2. During the model training, the total number of parameters was 2,587,201 out of which 328,705 were trainable. After passing through the Global Average pooling, the output of the MobileNetV2 was sent to the Dense layer. The provided input is convoluted by the dense layer with kernel size

1280×256 and bias 256. The batch normalization was then applied, and the ReLU as an activation function was employed in all layers until the final. The sigmoid function was employed in the last layer of the second dense block to categorize the images into two groups: COVID19(positive) and COVID19(negative) (see Fig. 13).

- XIII. **Stacking Model**: Stacking is an outstanding ensemble machine learning approach for predicting several nodes to create a new model with enhance model performance. Stacking allow to train numerous models to address similar tasks and then add their output to create a new model with superior performance. In this study, in the first phase twelve different transfer learning models (VGG-16, VGG-19, ResNet50, ResNet101, ResNet152, InceptionV3, InceptionResNet152, Xception, DenseNet121, DenseNet169, DenseNet201, and MobileNetV2) used for deep feature extractions from CT X-Ray COVID19 images. These deep features were flattened and concatenated by applying variety of hyper parameters such as adaptive learning rate, binary cross entropy, log loss and Adam optimizer. In the second phase eight well-known transfer learning models (VGG16, VGG19, Resnet50V2, Inception_v3, Inception_resnet_v2, Xception, Densenet121, Mobilenetv2) were stacked together to obtain the most accurate prediction from each model. The input image of $224 \times 224 \times 3$ was given to all twelve transfer learning models, which will calculate the feature maps of the image parallelly. These twelve models' features or predictions were then integrated and passed to the eight transfer learning models based on the majority voting algorithm. The deep features extracted by the deep neural models in the second layer by eight models were then integrated and passed to seven dense layers ahead to form a higher-level meta-model. Seven dense layers were added in the stacking model to

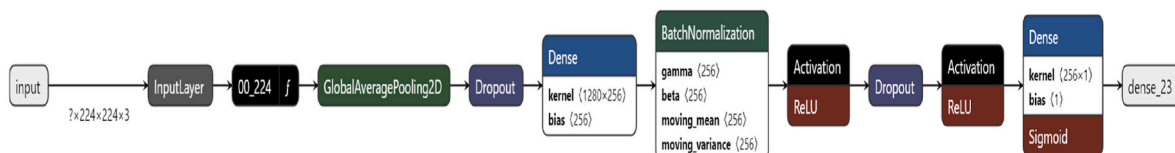


Fig. 13. MobileNetV2 architecture for COVID19 diagnosis in CXR images.

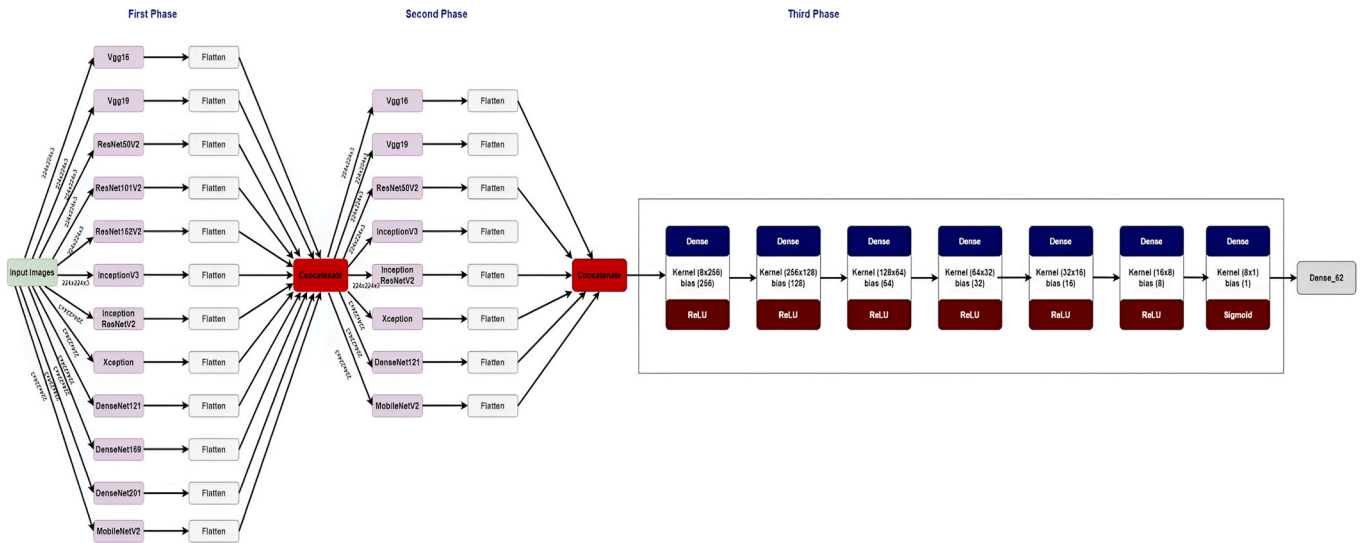


Fig. 14. Architecture of Stacking model for COVID-19 diagnosis in CXR images.

Table 2

Training hyper parameters used by the different transfer Learning models and Stacking model for the classification of COVID-19 in CXR images.

Hyperparameters	VGG16	VGG19	ResNet50V2	ResNet101V2	InceptionV3	DenseNet169	DenseNet201	Stacking Model
Optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
Learning rate	0.001	0.002	4e-5	4e-5	4e-5	3e-5	3e-5	1e-5
Batch size	32	32	32	32	32	32	32	32
Epochs	10	10	10	10	10	10	10	10
Loss	Binary Cross entropy	Binary Cross entropy	Binary Cross entropy	Binary Cross Entropy	Binary Cross entropy	Binary Cross entropy	Binary Cross Entropy	Binary Cross entropy
Classes	Binary	Binary	Binary	Binary	Binary	Binary	Binary	Binary
No of Trainable Parameters	132,097	132,097	525,313	525,313	525,313	427,009	492,545	103,880,841

Table 3

Performance metrics used for the evaluation of the model results.

Metric	Formula	Definition
Accuracy (Mahajan et al., 2024)	$\{(TP + TN)/(TP + TN + FP + FN)\}$	It is defined as the no. of images correctly classified to the total no of sample given.
Precision (Khan et al., 2022)	$\{TP/(TP + FP)\}$	It is defined as the ratio of the true predictions to the total no of predictions belonging to the positive class
Recall (Khan et al., 2022)	$\{TP/(TP + FN)\}$	It is defined as the ratio of the true predictions to the total no of predictions belonging to the positive and negative class
Matthews Correlation Coefficient (Chicco and Jurman, 2020)	$\{(TP*TN - FP*FN)/\sqrt{(TP + FP)(TN + FN)(TN + FP)(TN + FN)}\}$	It calculates the difference between the predicted number and the actual number
F1 Score (Mahajan et al., 2024)	$\{2 * (Precision * Recall)/(Precision + Recall)\}$	It is defined as the ratio of twice of the multiplication of precision and recall to the sum total of precision and recall
Jaccard Score (Niwattanakul et al., 2013)	$\{TP/(TP + FP + FN)\}$	It is defined as the ratio of the true predictions to the sum of predictions belonging to the positive and false negative class

TP: True Positive, TN: True Negative, FP: False Positive, FN: False Negative.

get the accurate features that will result in the classification of the COVID19 in CXR images with the highest accuracy among all other transfer learning models. As shown in Fig. 14, each dense layer has a kernel and bias of varying dimensions.

All dense layers used ReLU as an activation function except the last layer. The last layer, used sigmoid as an activation function for binary class classification. Table 2 give the detail of the different hyper parameters utilized in the training of the Stacking model and some other transfer learning models using COVID19 CXR dataset.

4. Results and discussions

This section focuses on the results calculated after testing several trained models on test data. The results were calculated based on different performance evaluation metrics (Taha and Hanbury, 2015; Liang, 2022) as shown in Table 3. After the pre-processing performed on the COVID19 CXR dataset was divided into train and test sets. The train set was used to train the various deep learning models explained in section 3, and testing was performed on the test set to classify them into two classes: those with COVID19 (positive and negative). All the deep learning models applied were pre-trained on the ImageNet dataset and incorporating transfer learning, the models training was done on the given dataset. Each model achieved above 90% accuracy in classifying the images into COVID19 (positive) and with COVID19 (negative). DenseNet201 achieved the highest accuracy Of 93.59% among other transfer learning models. The training graphs between the no. of epochs and each model's accuracy and loss are plotted and are displayed in Fig. 15. Table 4 contains the results of the twelve transfer learning

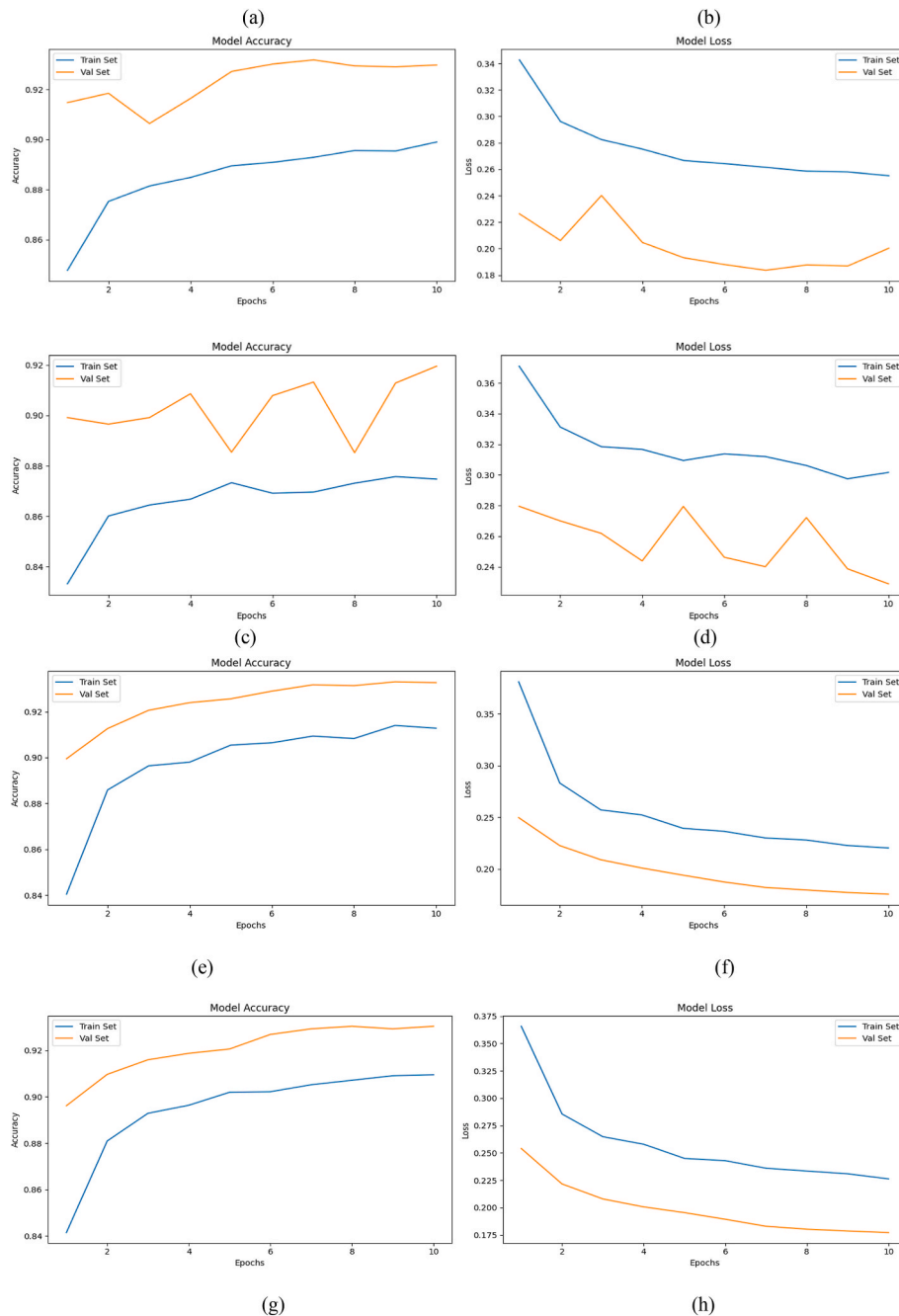


Fig. 15. Training Graphs plotted between model accuracy and no of epochs: (a) VGG16 (c) VGG19 (e) ResNet50 (g) ResNet101 (i) ResNet152 (k) InceptionV3 (m) InceptionResNetV2 (o) Xception (q) DenseNet121 (s) DenseNet169 (u) DenseNet201 (w) MobileNetV2 (y) Stacking model. Training Graphs plotted between model loss and no of epochs: (b) VGG16 (d) VGG19 (f) ResNet50 (h) ResNet101 (j) ResNet152 (l) InceptionV3 (n) InceptionResNetV2 (p) Xception (r) DenseNet121 (t) DenseNet169 (v) DenseNet201 (x) MobileNetV2 (z) Stacking model.

models calculated on the Test set using different performance metrics. The results of the proposed Stacking model calculated using different performance metrics on the test set are summarized in [Table 5](#).

Confusion Matrix: It is a matrix in which the accuracy of the predicted classes is analyzed ([Asuero et al., 2006](#)). A Confusion matrix is generally a 2x2 matrix in the case of binary class problems in which the model accuracy is visualized on Test data for the predicted values for which the actual values are already known. of the image classified by the different transfer learning model.

Correlation Matrix: A table representing correlation coefficients between the variables is called a Correlation matrix ([Aboutalebi et al.,](#)

[2022](#)). The correlation between the two variables is shown in each cell of the table. Correlation matrices are the summarization of data, also used as inputs for more advanced analyses and as diagnostics for those analyses.

The dark blue box in each of the different metrics in [Fig. 16](#) shows the True positive values or the true pixels of the image classified by the different transfer learning model in case of Confusion matrix and how much correlation is achieved between the actual and predicted pixel in case of Correlation matrix.

The training accuracy of the proposed model on the COID19 images achieved was 97.32%. The results of the proposed Stacking model

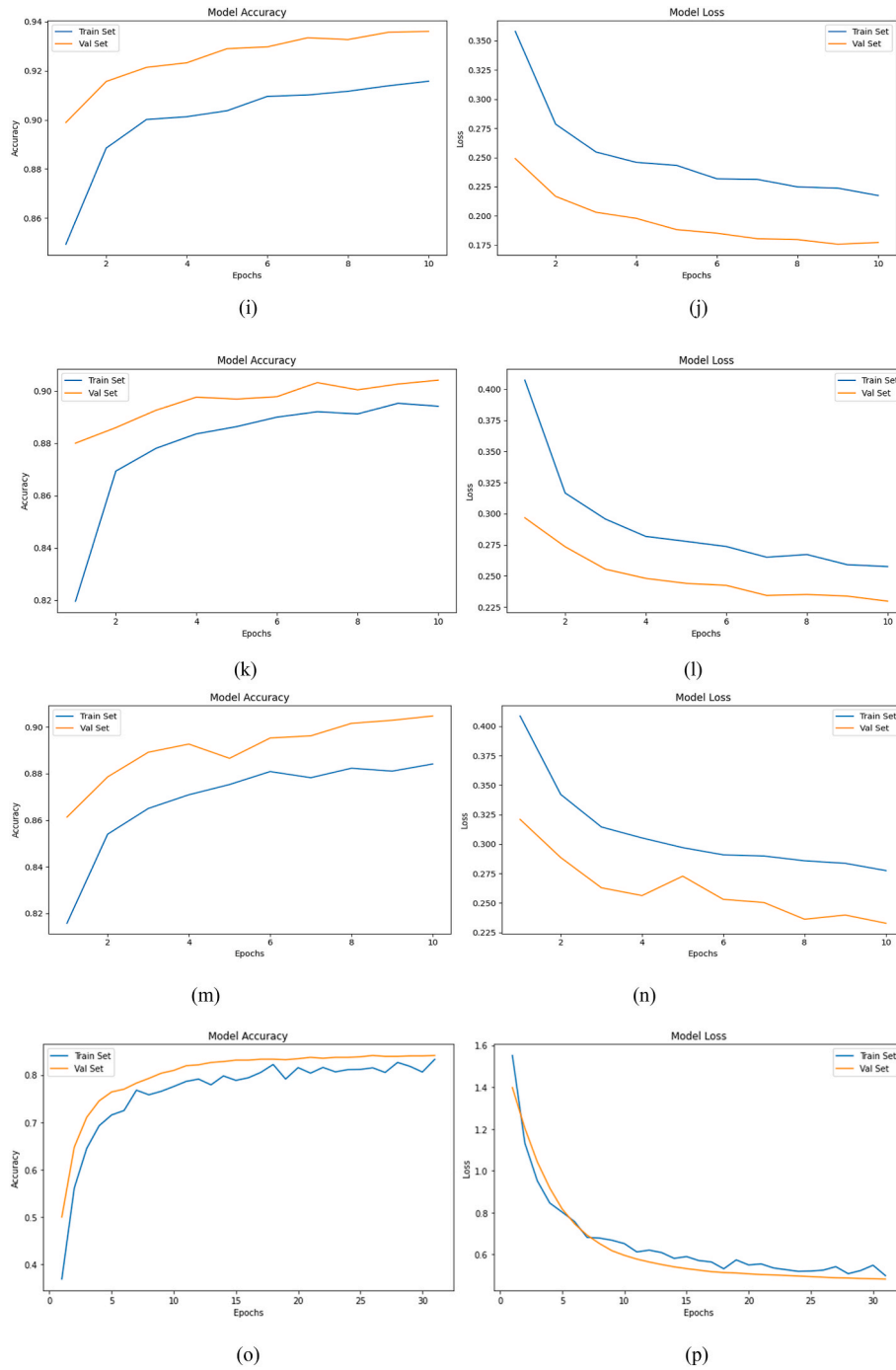


Fig. 15. (continued).

calculated on the accuracy parameter is compared with other state-of-the-art models implemented on the COVID19 CXR dataset as shown in Table 6. It is clear from Table 6 that our proposed model outperformed in classifying the COVID19 images and achieved the highest accuracy. The main reason behind the highest value of accuracy of our model is the three-layer stacking of transfer learning multimodel for the deep feature extraction from the input images. Also, the addition of the seven deep dense layers in the model for the further deep extraction of features and classification of the CXR images.

Transfer learning models performed better than others in the field of medical images. In this research, we utilized 12 transfer learning models to classify COVID19 (positive) and COVID19 (negative) on CXR dataset. Each model classified the COVID19 with an accuracy of greater than

90%. The findings indicate that automatic feature extraction and automatic feature identification from chest CT images, both of which are related to the diagnosis and categorization of the Covid19, may be considerably impacted by transfer learning using CNNs. By merging the output of various transfer learning models, a unique stacking model has been created to improve the feature extraction and classification of COVID-19. In first layer, a variety of transfer DNN models (VGG-16, VGG-19, ResNet50, ResNet101, ResNet152, InceptionV3, Inception-ResNet152, Xception, DenseNet121, DenseNet169, DenseNet201, and MobileNetV2) used for deep feature extractions from COVID19 CXR images. These deep features were flattened and concatenated by applying variety of hyper parameters such as adaptive learning rate, binarycrossentropy, log loss and Adam optimizer. The output is then

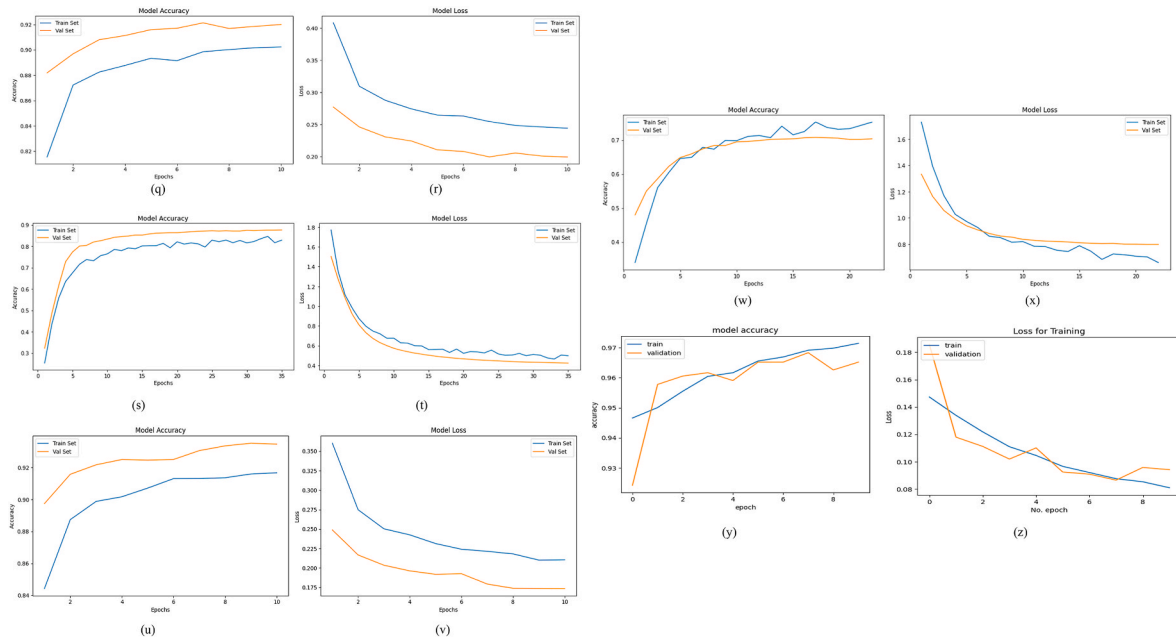


Fig. 15. (continued).

Table 4

Results calculated by different transfer learning models using various performance metrics on the test set.

Transfer Learning Models	Accuracy	Precision-weighted	Recall-weighted	F1-Score-weighted	Precision-macro	Recall-macro	F1-Score-macro	Jaccard_Score	Matthews_Correlation_Coefficient	Cohen_kappa_Score
DenselNet201	93.59	0.935972	0.935979	0.935969	0.935903	0.935631	0.93576	0.879667	0.871533	0.871521
DenselNet169	93.56	0.935647	0.935645	0.935628	0.935658	0.935203	0.935413	0.879068	0.87086	0.870828
ResNet152V2	93.03	0.930308	0.93031	0.930309	0.930107	0.930075	0.930091	0.869723	0.860182	0.860181
ResNet50V2	92.89	0.928974	0.928976	0.928975	0.928769	0.928737	0.928753	0.867396	0.857506	0.857506
DenselNet121	92.59	0.925966	0.925975	0.925967	0.925831	0.925633	0.925728	0.862169	0.851464	0.851457
ResNet101V2	92.09	0.920977	0.920974	0.920975	0.920715	0.920746	0.920731	0.853556	0.841462	0.841462
VGG16	91.83	0.92155	0.918306	0.917916	0.923499	0.915463	0.917447	0.848396	0.838924	0.835251
Xception	91.53	0.915626	0.915305	0.915348	0.914838	0.915675	0.915138	0.843931	0.830513	0.830298
InceptionV3	91.29	0.915013	0.912971	0.913037	0.913255	0.914507	0.912924	0.839995	0.827761	0.826029
VGG19	91.23	0.912301	0.912304	0.912302	0.912044	0.912013	0.912028	0.838784	0.824056	0.824056
InceptionResNetV2	90.29	0.903314	0.902968	0.903018	0.902488	0.903318	0.902779	0.823211	0.805805	0.805585
MobileNetV2	90.16	0.906131	0.901634	0.901016	0.90838	0.898217	0.900398	0.820048	0.806533	0.80142

Table 5

Results calculated by stacking model using various performance metrics on the test set.

Model	Accuracy	Precision-weighted	Recall-weighted	F1-Score-weighted	Precision-macro	Recall-macro	F1-Score-macro	Jaccard_Score	Matthews_CorrelationCoefficient	Cohen_kappa_Score
Stacking	0.9593	0.9593	0.9593	0.9593	0.9592	0.9590	0.9591	0.9218	0.9183	0.9183

sent to the DNN models (VGG-16, VGG-19, ResNet50, InceptionV3, InceptionResNet152, DenseNet121 and MobileNetV2). The deep features extracted by the deep neural models in the second layer were then integrated and passed to deep dense neural network model of seven layers to create a higher-level meta-model. Each layer in the seven dense layers used kernel and bias of varying dimensions. The problem of overfitting while training the model has been resolved through regularization of the layers of the transfer learning models and by using the dropout value of 0.01. It is clear from Table 4, Confusion Matrix, and Correlation coefficient matrix that the stacked model outperformed all other transfer learning models in use by having an accuracy rate on the provided dataset of 95.79% and also achieving highest value in all other performance metrics like precision, recall, F1_score etc. Stacking architecture allows for the simultaneous use of the features of many

models and produces output with improved outcomes.

5. Conclusion

This study focuses on classifying COVID-19 in CXR images into two categories: COVID-19 positive and COVID-19 negative, using various transfer learning models. A novel three-layer stacked transfer learning multimodal approach was proposed, outperforming state-of-the-art models in terms of accuracy and efficiency. The concatenated outputs from the stacked models were fed into seven deep dense layers with varying kernel and bias dimensions to enhance classification performance. The primary objective of this research was to develop a computationally efficient yet effective model for COVID-19 diagnosis and classification. The experiments were conducted on a large dataset of

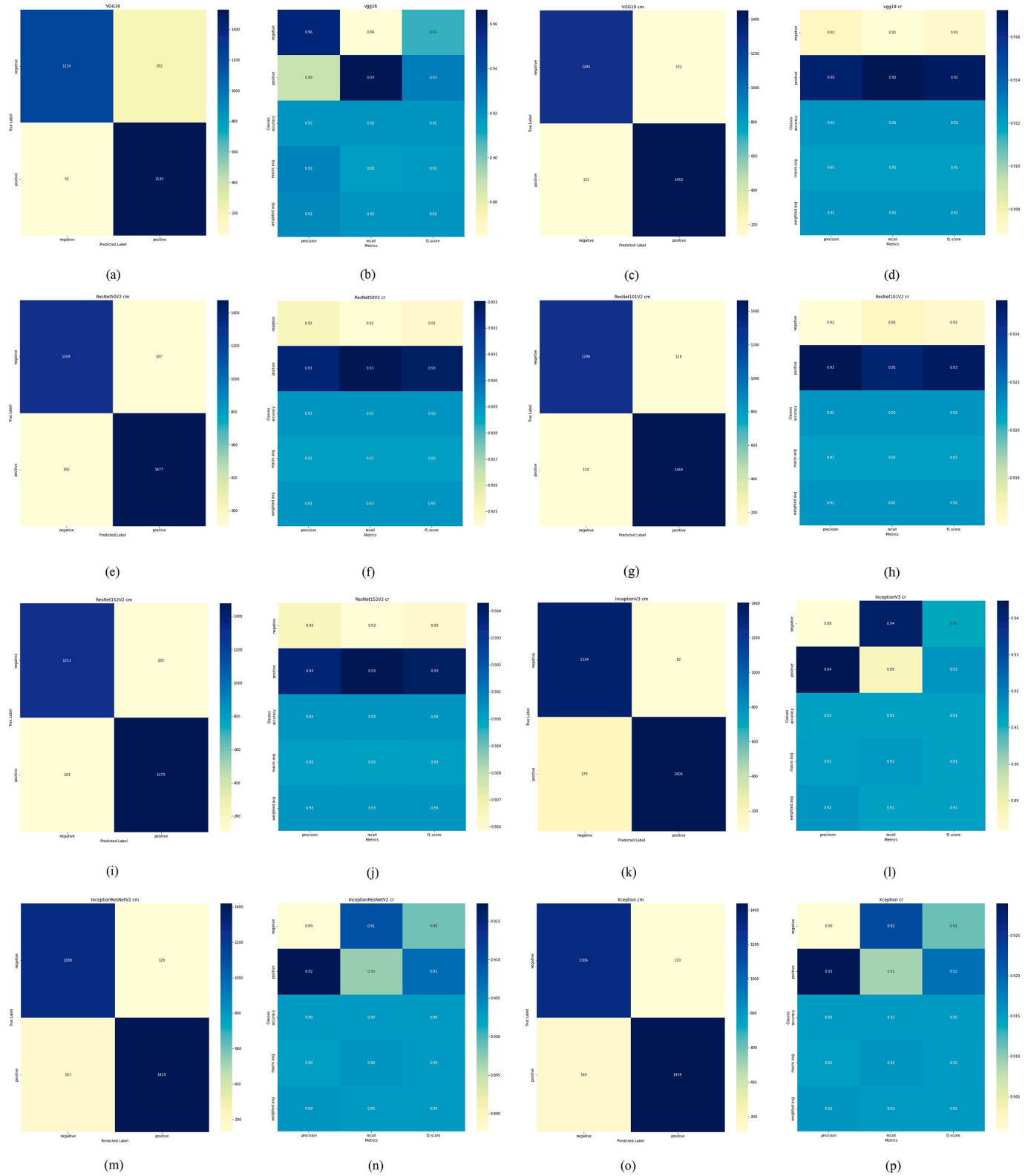


Fig. 16. Confusion matrix plotted between the actual pixels and the predicted pixels by the different transfer learning models: (a) VGG16 (c) VGG19 (e) ResNet50 (g) ResNet101 (i) ResNet152 (k) InceptionV3 (m) InceptionResNetV2 (o) Xception (q) DenseNet121 (s) DenseNet169 (u) DenseNet201 (w) MobileNetV2 (y) Stacking model. Correlation Coefficient matrix plotted between the actual pixels and the predicted pixels by the different transfer learning models: (b) VGG16 (d) VGG19 (f) ResNet50 (h) ResNet101 (j) ResNet152 (l) InceptionV3 (n) InceptionResNetV2 (p) Xception (r) DenseNet121 (t) DenseNet169 (v) DenseNet201 (x) MobileNetV2 (z) Stacking model.

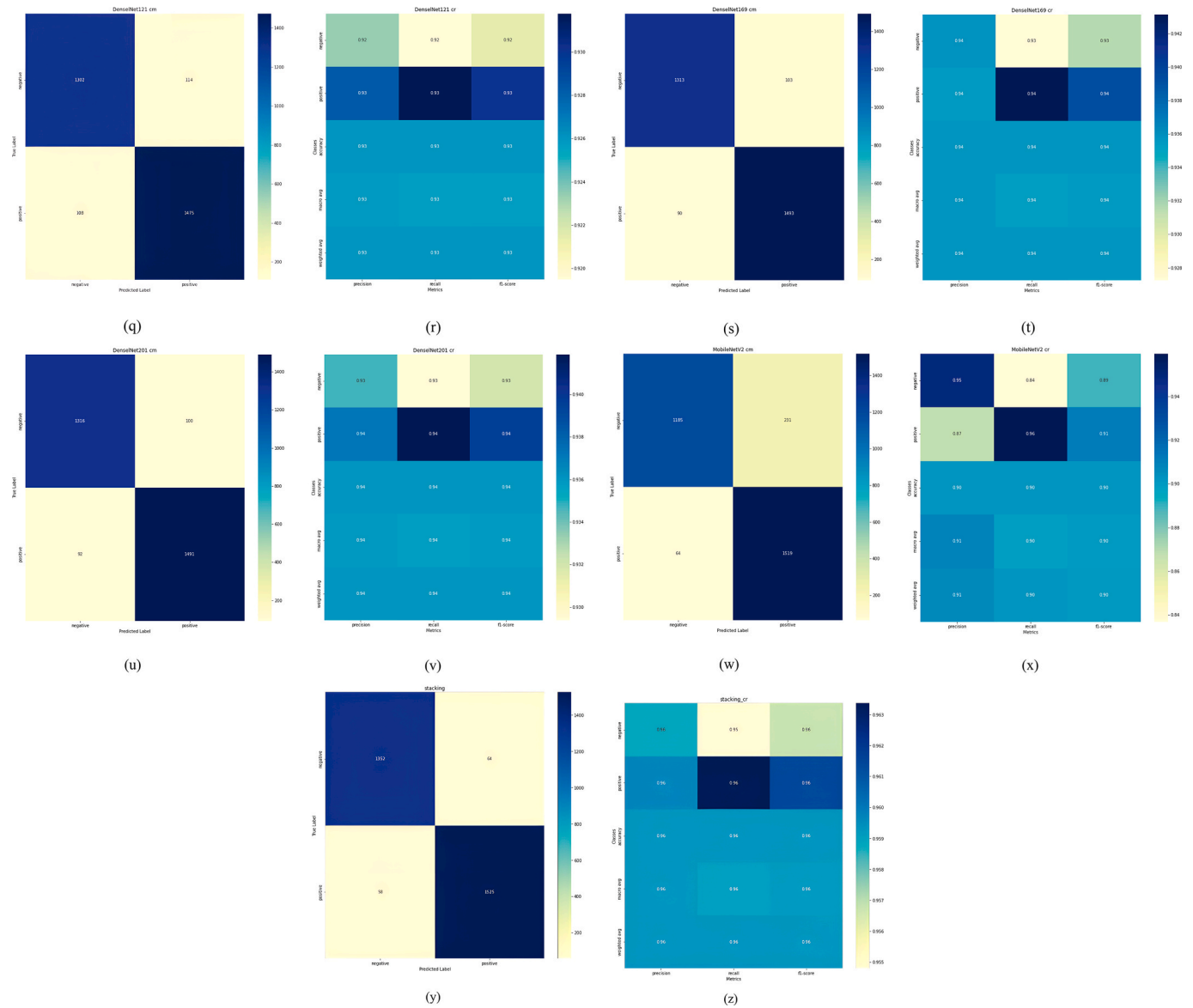


Fig. 16. (continued).

Table 6
Result comparison of the proposed Stacked multimodel with the state-of-the-art models.

References	Accuracy (%)
Proposed Stacking model	95.93
H. Aboutalebti et al. (Aboutalebti et al., 2022)	92.66
T. Szczepański et al. (Szczepański et al., 2022)	90.30
T. Ozturk et al. (Ozturk et al., 2020)	87.02

29,986 CXR images obtained from publicly available sources. This dataset included 13,992 COVID-19 negative and 15,994 COVID-19 positive images. For training purposes, 26,986 images were used, while 2999 images were reserved for model evaluation. The proposed model achieved an average accuracy of 95.79%, along with a precision of 95.44% and recall of 96.65%, demonstrating its effectiveness when compared to other state-of-the-art models applied to the same dataset. This study highlights the capability of the stacked transfer learning multimodal approach to handle high-volume datasets and deliver robust performance metrics. The findings indicate the potential of this

approach to assist in COVID-19 diagnosis with reduced computational overhead. Future research will focus on expanding the application area by classifying additional categories in COVID-19 images using real-time datasets. Moreover, segmentation techniques will be incorporated to identify affected areas in CXR images. Advanced deep learning models, such as UNET and ResNet, will be explored to improve the model's ability to detect and localize infection regions, ultimately aiding in better clinical decision-making and resource optimization.

CRedit authorship contribution statement

Bajjnath Kaushik: Project administration, Conceptualization. **Akshma Chadha:** Software, Resources. **Abhigya Mahajan:** Writing – review & editing. **Malvika Ashok:** Writing – original draft.

Ethical approval

No ethical approval was required.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The source of the Dataset used is mentioned in the manuscript.

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